Shane Sarosh

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Education

Vellore Institute of Technology (VIT) — Vellore, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.79/10 — *Aug. 2024 – May 2028*

Experience

The South Indian Bank Limited — Kochi, India
AI Intern — *May 2026 – June 2026*

- Built NoteForge, an AI-assisted QA platform that converts banking business-requirement documents into structured, traceable test cases using FastAPI, Pydantic, and LLM-based generation.
- Validated the system against digital-banking workflows spanning national eKYC identity verification (Aadhaar), core-banking integration (Finacle), and real-time payment rails (UPI/IMPS/NEFT/RTGS), including transaction limits and failure handling.

Research & Projects

Searchless Chess: Mechanistic Interpretability | *Python, JAX, Scikit-learn* — GitHub | Jun. 2026

- Probed eight chess concepts across all 17 layers of DeepMind's 270M-parameter Searchless Chess transformer, training 136 linear probes on 5,488 positions.
- Recovered board state with near-perfect fidelity from layer activations (material balance $R^2 = 0.999$, center control $R^2 = 0.989$, near-ceiling castling-rights AUC).
- Found that this interpretable structure peaks in layers 1–3 before degrading as later layers shift to encoding action values.
- Built a Chrome extension and WebSocket pipeline that visualizes the transformer's move preferences in real time during post-game analysis.

Hybrid Job Hunter | *Python, Playwright, GitHub Actions, Telegram API* — GitHub | Jul. 2026

- Engineered adapters for seven ATS and career platforms, combining structured APIs with Playwright browser automation to monitor JavaScript-rendered career pages on scheduled GitHub Actions runs with no dedicated infrastructure.
- Designed for fault tolerance — per-source health tracking, request budgets, and queued retries — running unattended on a scheduled cadence.

Adaptive Shortest-Path Framework (ASPF) | *C++, Python, Scikit-learn, NetworkX* — Feb. 2026

- Developed a runtime system that dynamically selects among BFS, Dijkstra's algorithm, and Dial's algorithm by analyzing graph structural properties.
- Trained a Random Forest classifier that correctly selects the fastest shortest-path algorithm on 88% of held-out test graphs.
- Improved execution time by up to 73% over fixed Dijkstra baselines on sparse graphs and 58% on uniform-weight graphs.

Technical Skills

Programming: Python, SQL, C++, JavaScript
Data Analysis: Pandas, NumPy, Matplotlib, exploratory data analysis, data cleaning, data visualization
Machine Learning: scikit-learn, JAX, model evaluation, classification, regression, linear probing
Development Tools: Git, FastAPI, Pydantic, Playwright, GitHub Actions

Certifications

Microsoft Certified: Azure AI Fundamentals | *Microsoft* — Credential
Qiskit Global Summer School 2026 - Quantum Excellence | *IBM* — *Aug. 2026*